

Enhanced Adversarial Drift Detection for MLOps Feature Monitoring

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Abstract—Machine learning models in production suffer from distribution drift, where evolving input features degrade performance. Existing unsupervised detectors signal that drift occurred, but not which features drifted or how severe the shift is. This paper proposes Enhanced Adversarial Drift Detection (EADD), extending the D3 discriminative detector with three contributions: (1) an Adaptive Sequential Permutation Test (ASPT) for rigorous drift confirmation, cutting permutation budget by 90–95%; (2) SHAP-based drift localisation and a Drift Severity Index (DSI) for feature-level diagnosis and severity quantification; and (3) benchmarking against seven unsupervised detectors on synthetic and real-world streams. EADD achieves full drift type coverage (4/4) with zero false alarms on synthetic data, detects drift on three of four annotated INSECTS datasets, and outperforms D3 on detection delay in all three detected cases (up to $8\times$ faster). SHAP attribution reaches perfect accuracy ($P@k = R@k = 1.0$) on controlled synthetic scenarios. On autocorrelated streams, EADD produces zero false alarms versus D3’s 54.6 ($p = 0.0101$, Mann–Whitney U).

Index Terms—Drift Detection, Machine Learning, MLOps, Feature Drift

I. INTRODUCTION

Machine learning models deployed in production environments encounter evolving data distributions, collectively termed distribution drift, that can degrade model performance if left undetected [?], [?]. Drift may manifest as changes in input features (covariate drift) or in the input–output relationship (concept drift); detecting such changes early is essential for maintaining reliable inference [?], [?].

The drift detection landscape spans three paradigms. *Supervised methods* such as DDM [?] monitor error rates but require timely ground-truth labels. *Statistical methods* employ hypothesis tests such as the Kolmogorov–Smirnov test [?], [?] but operate per feature, missing multivariate shifts. *Discriminative methods* such as D3 [?] train classifiers to distinguish reference from current data, achieving strong robustness in the benchmark by Lukats et al. [?]. However, all existing methods share a common limitation: they report only *whether* drift

occurred, not *which* features drifted or *how severe* the shift is.

D3 [?] exemplifies this limitation with two gaps: it applies no formal significance test (vulnerable to false alarms on autocorrelated data) and follows a “detect-and-discard” pattern, providing no feature-level attribution. Related work partially addresses these gaps: TRIPODD [?] enables feature-interaction testing but requires pre-specified interactions; HDPD [?] provides per-feature performance attributions but requires labels and operates post-hoc on batches.

In the unsupervised streaming literature, no single framework integrates (1) statistically rigorous multivariate detection, (2) feature-level attribution, and (3) severity quantification. Commercial platforms (Vertex AI [?], SageMaker Clarify [?]) address parts of this but operate in supervised or batch settings, not real-time streaming. EADD is an *integration contribution*: components draw on Gandy [?] for sequential testing and TreeSHAP [?], [?] for attribution, but their combination is non-trivial—the AUC gate must precede ASPT to bound Type I error, TreeSHAP must apply to a *discriminative* rather than predictive classifier, and severity must be dimensionality-invariant.

This paper proposes **Enhanced Adversarial Drift Detection (EADD)**, addressing this gap with three contributions:

- 1) Inspired by the statistical analysis of the classifier two-sample test [?], [?], we introduce an **Adaptive Sequential Permutation Test (ASPT)** [?] that gates drift confirmation via a formal permutation test, achieving 90–95% permutation savings while preserving Type I error control; H_{pred} is emitted as a continuous diagnostic output at every monitoring step (Section III-B,C).
- 2) Building on SHAP feature distribution analysis [?], we propose **SHAP-based drift localisation** and a **Drift Severity Index (DSI)** that combines detection confidence with attribution concentration into a continuous severity

score, identifying which features caused drift and quantifying how severe the shift is (Section III-D).

- 3) We benchmark EADD against seven unsupervised detectors on synthetic and real-world streams, evaluating detection coverage, false alarm control, and SHAP attribution accuracy (Section IV).

II. RELATED WORK

A. Drift Detection Methods

Drift detection methods can be classified along two axes: label availability and detection mechanism [?], [?].

Supervised methods monitor model error rates and require timely ground-truth labels. DDM [?] tracks the error rate and its standard deviation; EDDM [?] monitors inter-error distances, improving sensitivity to gradual drift. Both react only *after* performance has degraded, limiting utility when labels are delayed [?], [?].

Unsupervised methods operate without labels by directly monitoring data distributions. Statistical approaches include ADWIN [?], which dynamically adjusts sliding window sizes and can detect gradual drift, and KSWIN [?], which applies the Kolmogorov–Smirnov two-sample test over sliding windows for abrupt or sudden shifts. While statistically principled, these methods test each feature *independently* (univariate), making them blind to multivariate distributional shifts and causing multiple testing problems in high-dimensional settings [?].

Discriminative approaches frame drift detection as binary classification: an auxiliary classifier is trained to distinguish reference from current distributions [?], [?], [?], [?]. This naturally captures multivariate drift in a single test. The most prominent discriminative detector, **D3** [?], trains a logistic regression classifier on reference vs. current windows and triggers drift when AUC exceeds a fixed threshold. Lukats et al. [?] identified D3 as among the most robust unsupervised detectors across 29 real-world data streams. However, D3 has three weaknesses: (1) its linear discriminator misses non-linear feature interactions, causing failure on gradual and incremental drift; (2) it applies no formal significance test, relying on a fixed AUC threshold vulnerable to false alarms on autocorrelated data; and (3) it discards the trained classifier after computing AUC, providing no feature-level attribution. Other benchmarked detectors exhibit characteristic limitations: CSDDM [?] and IBDD [?] achieve broad coverage but with elevated false alarm rates; OCDD [?] and SPLL [?] are effective for specific drift types but miss others; BNDM struggles with high-dimensional data; and UDetect failed on 45% of scenarios [?].

B. Feature Attribution in Discriminative Detection

The *classifier two-sample test* (C2ST) [?] provides principled statistical foundations for discriminative detection: if a classifier reliably separates reference from current data (AUC well above 0.5), the distributions differ significantly. E-C2ST [?] strengthened statistical rigour via anytime-valid e-values. However, neither C2ST nor E-C2ST provides feature-level attribution or operates in a streaming setting. The prac-

tical need for such attribution was demonstrated at industrial scale by Pan et al. [?] at Uber, Qian et al. [?] for credit scoring, and by Google’s SHAP-based feature monitoring in Vertex AI [?].

Recent work has advanced individual aspects: TRIPODD [?] enables feature-interaction hypothesis testing, HDPD [?] provides per-feature performance decomposition, and Hinder et al. [?] formalised feature-wise drift definitions. However, no existing work systematically evaluates SHAP-based feature attribution accuracy within discriminative detection, nor integrates attribution with severity quantification in an unsupervised streaming pipeline.

III. PROPOSED METHODOLOGY

As identified in Section II, existing C2ST-based detectors lack continuous uncertainty monitoring, statistical significance testing, feature-level attribution, and severity quantification. EADD is a streaming pipeline for feature drift ($P(X)$ shift) detection that extends the C2ST framework [?] with three novel components to address these gaps (Fig. ??). The complete detection cycle is specified in Algorithm ??, which is referenced throughout this section; the four subsections that follow expand each step in turn.

Algorithm 1 EADD Detection Cycle

Require: Stream $\{x_t\}$, W_{ref} , W_{cur} , threshold τ , significance α

Ensure: Drift decision, H_{pred} , DSI, feature attribution, prescription

- 1: Update W_{cur} ; update W_{ref} via reservoir sampling [?]
 - 2: Train LightGBM on $W_{ref} \cup W_{cur}$ with labels $\{0, 1\}$
 - 3: Compute AUC via stratified 5-fold CV
 - 4: Compute H_{pred} via (??)
 - 5: **if** AUC $> \tau$ **then**
 - 6: $p \leftarrow \text{ASPT}(W_{ref}, W_{cur}, \text{AUC})$ {Adaptive early stopping}
 - 7: **if** $p < \alpha$ **then**
 - 8: Compute SHAP values via TreeSHAP
 - 9: Compute DSI via (??)
 - 10: Generate DSI-based prescription
 - 11: **return** DRIFT, H_{pred} , DSI, SHAP ranking, prescription
 - 12: **end if**
 - 13: **end if**
 - 14: **return** NO-DRIFT, H_{pred}
-

A. Step 1: Adaptive Reference Windowing

EADD maintains two windows: a *reference window* W_{ref} (N_{ref} samples) and a *current window* W_{cur} (N_{cur} samples). The reference window is updated by *reservoir sampling* [?]: each new sample x_t replaces a random element in W_{ref} with probability N_{ref}/t , which keeps an unbiased sample of historical data over long streams. The current window is a standard sliding buffer of recent observations. Monitoring is

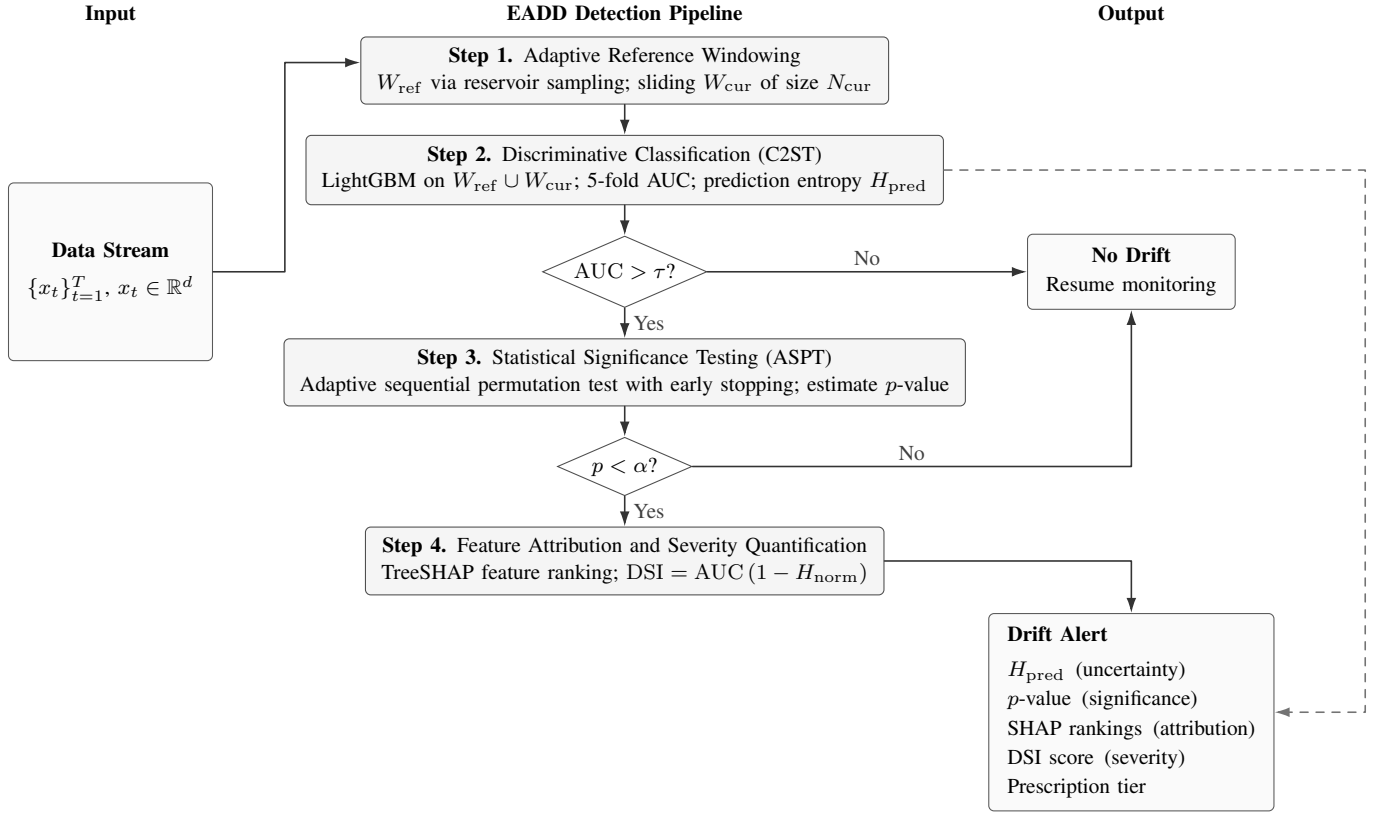


Fig. 1. EADD detection pipeline. W_{cur} is compared against W_{ref} via a discriminative classifier; H_{pred} provides a continuous uncertainty monitoring signal (dashed). $AUC > \tau$ triggers ASPT for statistical confirmation (Step 3); upon significance, TreeSHAP and DSI localise and quantify the shift (Step 4).

triggered every M samples to balance latency and cost; after confirmed drift, W_{ref} is reset to the new regime.

B. Step 2: Discriminative Classification and Prediction Entropy

Following the C2ST framework [?], EADD constructs a binary classification task: samples from W_{ref} are labelled 0 (reference) and samples from W_{cur} are labelled 1 (current). A LightGBM gradient boosting classifier [?] is trained on this combined dataset via stratified 5-fold cross-validation, and the out-of-fold AUC-ROC is computed. $AUC \approx 0.5$ indicates the classifier cannot distinguish the two windows (no drift); AUC above the threshold τ indicates the distributions are separable (drift suspected). We use LightGBM rather than D3's logistic regression [?] because gradient boosting captures non-linear feature interactions, providing a more powerful discriminator for the non-linear and incremental drift types that defeat D3's linear classifier.

Prediction Entropy. Beyond the aggregate AUC score, EADD extracts additional diagnostic information from the classifier's per-sample predictions. We compute the mean binary Shannon entropy of the discriminative classifier's predicted probabilities:

$$H_{pred} = \frac{-1}{n} \sum_{i=1}^n [\hat{p}_i \ln \hat{p}_i + (1 - \hat{p}_i) \ln(1 - \hat{p}_i)] \quad (1)$$

where $\hat{p}_i = f_{\theta}(x_i)$ is the probability that sample i belongs to W_{cur} : $H_{pred} \approx \ln 2 \approx 0.693$ indicates maximum uncertainty (no drift); $H_{pred} \approx 0$ indicates high classifier confidence (drift likely). Note: H_{pred} uses natural log; its maximum is $\ln 2$, not 1. H_{pred} remains informative at every monitoring step, including steps where $AUC < \tau$ and no detection gate is triggered, providing a continuous sub-threshold trend signal; its standalone predictive value is qualitative in the present work.

C. Step 3: Adaptive Sequential Permutation Test (ASPT)

If AUC exceeds a threshold τ , EADD applies an **adaptive sequential permutation test** inspired by Gandy [?]. Instead of a fixed budget of B permutations, ASPT monitors evidence accumulation and terminates early:

- 1) For each permutation $i = 1, \dots, B_{max}$: shuffle labels, retrain, compute $AUC_{perm}^{(i)}$. Update count $c_i = \#\{j \leq i : AUC_{perm}^{(j)} \geq AUC_{actual}\}$.
- 2) **Early reject H_0** : If $c_i = 0$, $i \geq B_{min}$, and $1/(i+1) < \alpha$, terminate and confirm drift with $p = 1/(i+1)$.
- 3) **Early accept H_0** : If $c_i/i > 2\alpha$ and $i \geq B_{min}$, terminate and declare no drift with $p = (c_i + 1)/(i + 1)$.
- 4) **Terminal**: If B_{max} is reached, $p = (c_{B_{max}} + 1)/(B_{max} + 1)$.

Drift is confirmed if $p < \alpha$. The two-stage gate is conservative: overall Type I error $\leq \Pr(AUC > \tau \mid H_0) \times \alpha \leq \alpha$. Under

$\alpha = 0.05$, early rejection requires $i \geq 20$ (since $1/21 \approx 0.048 < 0.05$); all confirmed detections in Experiments 1 and 3 terminated at exactly $i = 20$.

D. Step 4: Feature Attribution and Drift Severity Index

Upon drift confirmation, EADD exploits the trained discriminative classifier to diagnose *which* features drive the detected drift using SHAP (SHapley Additive exPlanations) [?].

Feature Attribution. EADD applies TreeSHAP [?] to the trained LightGBM discriminator. Per-feature importances $\bar{\phi}_j = \frac{1}{n} \sum_{i=1}^n |\phi_j^{(i)}|$ are normalised into a contribution vector $\mathbf{s} = (s_1, \dots, s_d)$, $s_j = \bar{\phi}_j / \sum_k \bar{\phi}_k$, identifying which features drive the shift and feeding the DSI below.

Drift Severity Index (DSI). The normalised contribution vector $\mathbf{s}$ computed from TreeSHAP directly feeds the DSI: a composite severity metric combining detection confidence with attribution concentration:

$$\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s})) \quad (2)$$

where $H_{\text{norm}}(\mathbf{s}) = -\sum_{j=1}^d s_j \log_2 s_j / \log_2 d$ is the normalised Shannon entropy [?]. The $\log_2 d$ denominator ensures $H_{\text{norm}} \in [0, 1]$ regardless of dimensionality. DSI-based prescriptions map severity to recommended action. The tier boundaries below are proposed heuristics requiring domain-specific calibration; they are not empirically validated in the present work (all experiment DSI values fall in the Low tier):

- **Critical (DSI > 0.6):** Concentrated drift, audit specific data pipeline.
- **Moderate (0.3–0.6):** Subset drift, investigate shared data source.
- **Low (< 0.3):** Diffuse shift, schedule full model retraining.

IV. EXPERIMENTAL EVALUATION

A. Experimental Design

Three experiments were designed to validate EADD’s detection accuracy, feature attribution, and computational efficiency. Following the evaluation protocol of Lukats et al. [?], EADD was benchmarked against seven unsupervised detectors: D3 [?], BNDD, CSDDM [?], IBDD [?], OCDD [?], SPLL [?], and UDetect.

Datasets. Table ?? summarises the datasets used across experiments.

Experiment 1 (Synthetic Data): Synthetic streams ($n=5,000$, $d=5$) with controlled abrupt, gradual, incremental, and recurring drift. All eight detectors were compared on each drift type. Additionally, false alarm robustness was evaluated on four stable synthetic streams (Gaussian i.i.d., autocorrelated AR(1) with $\phi=0.8$, heteroscedastic, correlated with $\rho=0.7$) containing zero drift, with five independent runs per stream type.

Experiment 2 (Real-World Benchmark): Thirteen datasets (Table ??): four INSECTS variants with annotated drift points [?] are evaluated by Recall and Mean Time to Detection (MTD); nine unannotated streams [?] are evaluated

TABLE I
DATASETS USED IN EXPERIMENTS (†=ANNOTATED; DRIFT COUNT IN PARENTHESES; UNANNOTATED = NO GROUND-TRUTH DRIFT-POINT LABELS).

Dataset	n	d	Drift?
<i>Synthetic (Experiments 1, 3)</i>			
Temporal Patterns	5K	5	Yes (4 types)
Feature Attribution	10K	10	Yes (3 scenarios)
<i>Experiment 2: Real-World [?], [?]</i>			
INSECTS Abrupt Bal.†	52.8K	33	Yes (5)
INSECTS Gradual Bal.†	24.2K	33	Yes (1)
INSECTS IncrAbrupt Bal.†	80.0K	33	Yes (2)
INSECTS IncrRecur Bal.†	80.0K	33	Yes (2)
Electricity	45.3K	8	Unannotated
INSECTS Incremental	57.0K	33	Unannotated
Luxembourg	1.9K	31	Unannotated
NOAAWeather	18.2K	8	Unannotated
OutdoorObjects	4.0K	21	Unannotated
Ozone	2.5K	72	Unannotated
PokerHand	100K	10	Unannotated
Powersupply	29.9K	2	Unannotated
RialtoBridge	82.3K	27	Unannotated

by total detection count (selectivity). Only EADD and D3 are compared in this experiment.

Experiment 3 (SHAP Attribution Accuracy): Three controlled synthetic scenarios ($n=10,000$, $d=10$ features, drift injected at $t=5,000$): (a) univariate drift in F3 only (mean shift $+2\sigma$), (b) subset drift in F2/F5/F7 simultaneously, (c) multivariate drift in all features. Precision@k (P@k) and Recall@k (R@k) measure whether the top- k SHAP-ranked features exactly match the k truly drifted features.

Configuration. EADD: LightGBM (50 estimators, lr 0.1, 31 leaves), $N_{\text{ref}}=500$, $N_{\text{cur}}=200$, $M=50$, ASPT $B_{\text{max}}=199$, $B_{\text{min}}=10$, $\alpha=0.05$, $\tau=0.7$. D3: logistic regression (liblinear), matched windows, $\tau=0.7$. Python 3.10, LightGBM 4.1.0, SHAP 0.42.1, scikit-learn 1.3.2; fixed seeds across detectors.

B. Experiment 1: Synthetic Data

Table ?? presents drift type coverage and false alarm rates for all eight detectors.

TABLE II
EXPERIMENT 1: SYNTHETIC DRIFT DETECTION RESULTS. TYPES = NUMBER OF DRIFT PATTERNS DETECTED OUT OF 4. MEAN FA = AVERAGE FALSE ALARMS PER TYPE. LIFT = DETECTIONS PER TRUE DRIFT POINT, FOLLOWING LUKATS ET AL. [?] REPORTING.

Detector	Types	Coverage	Mean FA	Lift
EADD	4/4	100%	0.00	1.00
BNDD	4/4	100%	4.75	5.75
CSDDM	4/4	100%	5.50	6.50
IBDD	4/4	100%	46.25	47.25
OCDD	4/4	100%	17.50	18.50
D3	2/4	50%	0.00	1.00
SPLL	2/4	50%	0.00	1.00
UDetect	0/4	0%	n/a	n/a

EADD is the only detector that achieved full drift-type coverage (4/4) with zero false alarms. We attribute the coverage

advantage to LightGBM’s non-linear discrimination, consistent with D3’s documented failure on gradual and incremental drift [?], [?], though no ablation isolates this from other design choices (see Section VI). The ASPT then eliminates spurious detections on stable streams, bringing false alarms to zero. BNDD, CSDDM, IBDD, and OCDD also achieved 4/4 coverage (through different discriminators) but with substantially higher false alarm costs (4.75, 5.50, 46.25, and 17.50 mean FA, respectively), demonstrating that coverage and false alarm control are separable objectives. D3 and SPLN detected only 2/4 types. UDetect failed on all four types.

False Alarm Robustness. On four stable streams EADD has zero false alarms; D3 is vulnerable to autocorrelation (87.4 at $\tau=0.6$; 54.6 at $\tau=0.7$). Mann–Whitney U: significant at $\tau=0.6$ ($p=0.0101$, D3 benchmark default [?]); not significant at the matched $\tau=0.7$ ($p=0.2266$) although EADD still produces 0 vs. 54.6.

C. Experiment 2: Real-World Benchmark

Table ?? presents results on the four annotated INSECTS datasets. Both detectors missed InsectsGradual, a slow transition detectable only at higher false alarm cost [?]. On all three jointly detected datasets, EADD outperforms D3 on MTD, most markedly on incremental drift types: InsectsIncrAbrupt MTD 33 vs. 272.5 ($8\times$ faster) and InsectsIncrRecur MTD 133 vs. 323.5 ($2.4\times$ faster). On InsectsAbrupt, EADD achieves MTD 135 vs. 155.5 (13% faster) with lower recall (0.60 vs. 0.80), reflecting its precision-first operating point. All MTD values are single-run point estimates. On nine unannotated datasets, EADD and D3 produced identical total detection counts (467 each).

TABLE III

EXPERIMENT 2: ANNOTATED REAL-WORLD DATASETS. PREC. = PRECISION; REC. = RECALL ON ANNOTATED DRIFT POINTS. MTD = MEAN TIME TO DETECTION (SAMPLES; LOWER IS BETTER). “N/A” = MISSED. LOW PRECISION ON INCREMENTAL STREAMS REFLECTS SPARSE ANNOTATION OF GRADUAL TRANSITIONS [?].

Dataset	Metric	EADD	D3
InsectsAbrupt	Precision	0.33	0.44
InsectsAbrupt	Recall	0.60	0.80
InsectsAbrupt	MTD	135	155.5
InsectsGradual	Precision	0.00	0.00
InsectsGradual	Recall	0.00	0.00
InsectsGradual	MTD	n/a	n/a
InsectsIncrAbrupt	Precision	0.14	0.22
InsectsIncrAbrupt	Recall	1.00	1.00
InsectsIncrAbrupt	MTD	33	272.5
InsectsIncrRecur	Precision	0.17	0.18
InsectsIncrRecur	Recall	1.00	1.00
InsectsIncrRecur	MTD	133	323.5

D. Experiment 3: SHAP Attribution Accuracy

Table ?? presents the SHAP attribution accuracy using precision@k and recall@k metrics.

EADD achieved perfect attribution accuracy across all three scenarios; the DSI-based prescription was correct in all cases. The multivariate scenario ($k=d=10$) trivially satisfies

TABLE IV

EXPERIMENT 3: SHAP ATTRIBUTION ACCURACY ON CONTROLLED SYNTHETIC STREAMS ($n=10,000$, $d=10$ FEATURES; DRIFT INJECTED AT $t=5,000$ IN F3 ONLY / F2+F5+F7 / ALL FEATURES). ALL SCENARIOS DETECTED AT STEP 5,149. DSI = $AUC \times (1 - H_{\text{NORM}}(\mathbf{s}))$; HIGHER DSI INDICATES MORE CONCENTRATED, SEVERE DRIFT. RX (✓) = CORRECT DSI-BASED DRIFT-TYPE CLASSIFICATION.

Scenario	AUC	P@k	R@k	Top (%)	DSI	Rx
Univariate (1/10)	0.784	1.0	1.0	F3 (52.3)	0.193	✓
Subset (3/10)	0.809	1.0	1.0	F5 (24.5)	0.096	✓
Multivariate (10/10)	0.806	1.0	1.0	F2 (14.6)	0.016	✓

$P@k=R@k=1.0$ for any ranking, so the meaningful result is the univariate case (1/10 features drifted, correct feature F3 ranked first at 52.3%). DSI correctly orders scenarios by concentration (univariate > subset > multivariate), as algebraically expected from (??). All DSI values fall in the “Low” tier (< 0.3) under moderate $+2\sigma$ shifts; Critical and Moderate tiers require more severe drift and are not empirically reached here.

E. Component Ablation, Runtime, and Correlated Features

To isolate whether EADD’s gains over D3 stem from the LightGBM upgrade or from ASPT, we ran an ablation that disables ASPT and confirms drift whenever $AUC > \tau$ (denoted EADD_{–ASPT}). Table ?? reports per-stream wall-clock time as feature dimension d varies (fixed $N_{\text{cur}}=200$, $n=8000$, drift at $t=4000$, $+2\sigma$ shift, single Apple M-series CPU; LightGBM 4.1.0).

TABLE V

WALL-CLOCK SCALING AND COMPONENT ABLATION ($n=8000$, SINGLE APPLE M-SERIES CPU; LIGHTGBM 4.1.0). TOP: FEATURE DIMENSION d AT $N_{\text{cur}}=200$; COVERAGE ON THE FOUR SYNTHETIC DRIFT TYPES. BOTTOM: WINDOW SIZE N_{cur} AT $d=10$.

(a) Vary d ($N_{\text{cur}}=200$)				
Variant	$d=10$	$d=20$	$d=50$	Cov.
EADD-Full	6.93s	10.76s	23.11s	4/4
EADD _{–ASPT}	6.68s	10.43s	22.64s	4/4
D3	0.07s	0.09s	0.14s	2/4
(b) Vary N_{cur} ($d=10$)				
Variant	$N=100$	$N=200$	$N=500$	$N=1000$
EADD-Full	5.83s	6.91s	10.52s	11.22s
EADD _{–ASPT}	5.62s	6.68s	10.23s	10.58s
D3	0.11s	0.07s	0.04s	0.03s

Component contributions and runtime. EADD_{–ASPT} retains 4/4 synthetic coverage (Table ??a), so the coverage gain over D3 stems from the LightGBM upgrade, not the ASPT gate; ASPT’s role is significance control on autocorrelated streams (suppressing 54.6 spurious detections at $\tau=0.7$). EADD-Full and EADD_{–ASPT} differ by $< 5\%$ across all d and N , confirming the 90–95% ASPT saving is realised *within* the significance stage and is orthogonal to the LightGBM-driven $\sim 80\text{--}160\times$ overhead vs. D3. The dominant cost is per-step LightGBM training: each monitoring step at $M=50$ refits a 50-tree boosted model on $N_{\text{ref}}+N_{\text{cur}}$ samples (700 at default),

giving ~ 160 trainings per 8K-sample stream; this scales near-linearly in d (Table ??a) and sub-linearly in N (Table ??b: N 200 $\rightarrow$ 1000 raises time only $1.6\times$).

Correlated features. On Cholesky-mixed streams ($d=10$, $n=8000$, $+2\sigma$ shift in 3 features), EADD first-detects at $t=4099$ for $\rho \in \{0, 0.5, 0.9\}$ (D3 at $t=4117$); stable correlated streams give zero false alarms. SHAP under correlation should be read at feature-group granularity [?].

V. CONCLUSION

EADD extends discriminative drift detection with ASPT+ H_{pred} for rigorous confirmation, SHAP+DSI for feature-level diagnosis and severity, and benchmarking against seven detectors. Synthetic coverage 4/4 (LightGBM delivers coverage, ASPT controls false alarms); MTD beats D3 on all three jointly detected INSECTS streams (up to $8\times$); SHAP $P@k=R@k=1.0$ in the univariate case, robust to $\rho \in \{0, 0.5, 0.9\}$; 0 false alarms vs. D3's 54.6 on autocorrelated streams ($\tau=0.7$). EADD answers *whether, when, which, and how severe* in one pipeline.

VI. LIMITATIONS AND FUTURE WORK

EADD's $\sim 50\text{--}160\times$ wall-clock overhead vs. D3 is dominated by LightGBM training; ASPT's 90–95% saving is orthogonal. DSI ordering is algebraic from (?); external validity is unestablished and tiers need domain calibration. SHAP attributions are correlational, not causal [?], and prescriptions are heuristic. **Multiple testing over time:** a test runs every $M=50$ samples ($\sim 75\%$ overlap, $\alpha=0.05$) without correction across ~ 600 tests on a 30K-sample stream; Bonferroni gives $\alpha' \approx 8.3 \times 10^{-5}$, below ASPT's $1/(B_{\text{max}}+1)=0.005$ floor, so strict FWER is not achieved—the zero-false-alarm result is empirical, not guaranteed. Future work: anytime-valid e-values [?]; causal attribution [?]; adversarial robustness; systematic study of DSI distribution and tier-trigger frequency across diverse production datasets to ground the heuristic boundaries empirically.

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REFERENCES

- [1] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang, “Learning under concept drift: A review,” *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 12, pp. 2346–2363, 2019.
- [2] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, “A survey on concept drift adaptation,” *ACM Comput. Surv.*, vol. 46, no. 4, pp. 1–37, 2014.
- [3] G. I. Webb, R. Hyde, H. Cao, H. L. Nguyen, and F. Petitjean, “Characterizing concept drift,” *Data Min. Knowl. Discov.*, vol. 30, no. 4, pp. 964–994, 2016.
- [4] F. Bayram, B. S. Ahmed, and A. Kassler, “From concept drift to model degradation: An overview on performance-aware drift detectors,” *Knowl.-Based Syst.*, vol. 245, 108632, 2022.
- [5] G. Widmer and M. Kubat, “Learning in the presence of concept drift and hidden contexts,” *Mach. Learn.*, vol. 23, no. 1, pp. 69–101, 1996.
- [6] S. Rabanser, S. Günnemann, and Z. Lipton, “Failing loudly: An empirical study of methods for detecting dataset shift,” in *Proc. NeurIPS*, 2019, pp. 1396–1408.

- [7] C. Raab, M. Heusinger, and F.-M. Schleif, “Reactive soft prototype computing for concept drift streams,” *Neurocomputing*, vol. 416, pp. 340–351, 2020.
- [8] J. Gama, P. Medas, G. Castillo, and P. Rodrigues, “Learning with drift detection,” in *Proc. Brazilian Symp. Artif. Intell.*, 2004, pp. 286–295.
- [9] Ö. Gözüaık, A. Büyükçakır, H. Bonab, and F. Can, “Unsupervised concept drift detection with a discriminative classifier,” in *Proc. ACM CIKM*, 2019, pp. 2365–2368.
- [10] D. Lukats, O. Zielinski, A. Hahn, and F. Stahl, “A benchmark and survey of fully unsupervised concept drift detectors on real-world data streams,” *Int. J. Data Sci. Analytics*, 2024, doi: 10.1007/s41060-024-00620-y.
- [11] M. Baena-García et al., “Early drift detection method,” in *Proc. ECML PKDD Workshop on Knowledge Discovery from Data Streams*, 2006, pp. 77–86.
- [12] A. Bifet and R. Gavaldà, “Learning from time-changing data with adaptive windowing,” in *Proc. SIAM Int. Conf. Data Mining*, 2007, pp. 443–448.
- [13] J. Pan, V. Pham, M. Dorairaj, H. Chen, and J.-Y. Lee, “Adversarial validation approach to concept drift problem in user targeting automation systems at Uber,” in *Proc. AdKDD Workshop, KDD*, 2020.
- [14] H. Qian, B. Wang, P. Ma, L. Peng, S. Gao, and Y. Song, “Managing dataset shift by adversarial validation for credit scoring,” *arXiv preprint arXiv:2112.10078*, 2021.
- [15] J. S. Wan and S.-D. Wang, “Concept drift detection based on pre-clustering and statistical testing,” *J. Internet Technol.*, vol. 22, no. 2, pp. 275–286, 2021.
- [16] V. M. A. Souza, A. R. S. Parmezan, F. A. Chowdhury, and A. Mueen, “Efficient unsupervised drift detector for fast and high-dimensional data streams,” *Knowl. Inf. Syst.*, vol. 63, no. 6, pp. 1497–1527, 2021.
- [17] Ö. Gözüaık and F. Can, “Concept learning using one-class classifiers for implicit drift detection in evolving data streams,” *Artif. Intell. Rev.*, vol. 54, no. 5, pp. 3725–3747, 2021.
- [18] L. I. Kuncheva, “Change detection in streaming multivariate data using likelihood detectors,” *IEEE Trans. Knowl. Data Eng.*, vol. 25, no. 5, pp. 1175–1180, 2013.
- [19] D. Lopez-Paz and M. Oquab, “Revisiting classifier two-sample tests,” in *Proc. ICLR*, 2017.
- [20] A. Taly, K. Sato, and C. Gruia, “Monitoring feature attributions: How Google saved one of the largest ML services in trouble,” *Google Cloud Blog*, 2021.
- [21] AWS, “Amazon SageMaker Clarify: Model monitoring,” Amazon Web Services Documentation, 2025.
- [22] J. S. Vitter, “Random sampling with a reservoir,” *ACM Trans. Math. Softw.*, vol. 11, no. 1, pp. 37–57, 1985.
- [23] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Proc. NeurIPS*, 2017, pp. 3149–3157.
- [24] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Proc. NeurIPS*, 2017, pp. 4766–4777.
- [25] V. M. A. Souza, D. M. Dos Reis, A. G. Maletzke, and G. E. A. P. A. Batista, “Challenges in benchmarking stream learning algorithms with real-world data,” *Data Min. Knowl. Discov.*, vol. 34, pp. 1805–1858, 2020.
- [26] T. Pandeva, T. Bakker, C. A. Naesseth, and P. Forré, “E-evaluating classifier two-sample tests,” *Trans. Mach. Learn. Res.*, 2024.
- [27] P. Panda, S. S. Kancheti, V. N. Balasubramanian, and G. Sinha, “Interpretable model drift detection,” in *Proc. ACM IKDD CODS-COMAD*, 2024.
- [28] H. Singh, F. Xia, A. Subbaswamy, A. Gossmann, and J. Feng, “A hierarchical decomposition for explaining ML performance discrepancies,” in *Proc. NeurIPS*, 2024.
- [29] A. Gandy, “Sequential implementation of Monte Carlo tests with uniformly bounded resampling risk,” *J. Am. Stat. Assoc.*, vol. 104, no. 488, pp. 1504–1511, 2009.
- [30] S. M. Lundberg, G. Erion, H. Chen et al., “From local explanations to global understanding with explainable AI for trees,” *Nature Mach. Intell.*, vol. 2, pp. 56–67, 2020.
- [31] F. Hinder, V. Vaquet, and B. Hammer, “Feature-based analyses of concept drift,” *Neurocomputing*, vol. 600, 127968, 2024.
- [32] K. Budhathoki, D. Janzing, P. Bloebaum, and L. Minorics, “Why did the distribution change?” in *Proc. AISTATS*, 2021, pp. 1666–1674.
- [33] C. E. Shannon, “A mathematical theory of communication,” *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, 1948.